

Test Statistics

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Least Squares, Tests

- Monday: Least Squares: the math behind that call to fit. ✓
- Today: [Test Statistics](#) (χ^2 , Z, T, F/ ANOVA, LR, AIC, BIC)

Test Statistics

Test Statistics are Random Variables based on data. They allow us to calculate probabilities, often in the form of p-values.

In Least Squares Estimation, we used the **Test Statistic** χ^2 to estimate the model parameters, and their uncertainties, and **we can also use its value directly to tell us how good our fit is.**

In Maximum Likelihood Estimation, we used the **Test Statistic** $\log \mathcal{L}_{\hat{\theta}}$ to estimate the model parameters, but the value of $\log \mathcal{L}_{\hat{\theta}}$ tells us nothing about how well the model fits the data. The more data we use, the larger $\log \mathcal{L}_{\hat{\theta}}$ will be. **We need to compare log likelihoods to check our model against another.**

Hypothesis Testing Recap

A **Hypothesis** is an educated guess at the explanation for some occurrence. It must be based on real observations and it must be testable.

- The **Null Hypothesis** H_0 is the educated guess that the most obvious (boring?) explanation for the data is the correct one.
- The **Alternative Hypothesis** H_1 is an alternative to H_0 .

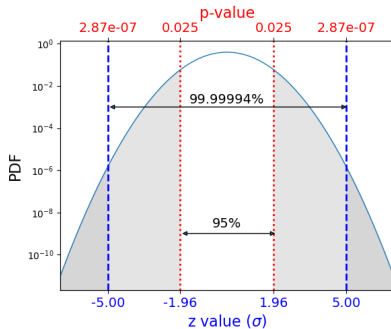
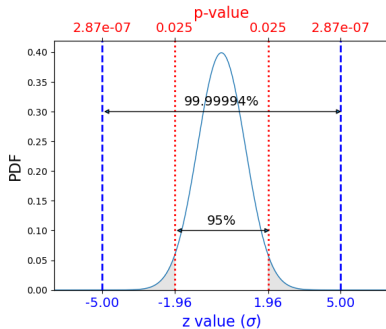
We need to put in place **Strict Criteria** to prevent us from manipulating our results. This means fully defining the test before looking at the data.

- A **Critical Region** W is a region of the Sample Space in which the probability of finding the data, given the Null Hypothesis, is very small. We quantify 'very small' using a **Significance Level**, α , with a typical value of $\alpha = 0.05$ (5%).

A **Positive Result** means that we have found our data x in W , and so we **reject the Null Hypothesis**. We will have a false positive 5% of the time if we use the popular significance level $\alpha = 0.05$.

P-values and Z-values recap

Recall the Z-value and p-value are different ways of measuring the same thing:



P-values for a fit

Recall the Z-value and p-value are different ways of measuring the same thing:

$$P(|Z| \leq 3) = 99.73\%, \text{ z-value is } 3\sigma, \text{ p-value is } 0.0027.$$

A p-value of 0.0027 indicates that, given H_0 , we would expect to observe this value for the data in 0.27% of repeated experiments. A small p-value (large Z-value) is an indicator of rarity that is interpreted as 'highly significant'.

When we think about probabilities in terms of fitting, our aim is to have the fit match the data as closely as possible, such that we can be confident in the parameter estimates.

So, when we calculate the p-value for a fit, we do not find small p-values exciting! A very small (or very large!) p-value for a fit means that we are very unlikely to have done a good job.

Degrees of Freedom

The number of degrees of freedom are central to many statistical tests.

Number of Degrees of Freedom: $\nu = N - N_p - N_c$

N : Number of measurements, aka sample size.

N_p : Number of parameters.

N_c : Number of constraints.

Example 1: I have a dataset of 10 measurements, with a mean of 1.63. How many degrees of freedom are there?

Example 2: In a Least Squares fit I am estimating the parameters m and c of a straight line. I have 20 data points. How many degrees of freedom are there?

Case Studies

Normal Tests (Z-test, T-test, F-test):

- 1 Is the mean height of DAT students equal to the mean UK height?
- 2 Is the mean height of Mechanics students equal to that of DAT students?

Model Tests (χ^2 , LLR, AIC, BIC):

- 1 My friend suggested a more complicated model might better fit my data. How do I prove him wrong?
- 2 What if he suggests a completely different model?

The Z Test

Example usage: test if the sample mean $\bar{x}$ equals a hypothesised mean μ_0 .

Restrictions: Data are $X \sim \text{Norm}(\mu, \sigma)$, True Variance σ^2 is known.

$$z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{N}}$$

The denominator $\sigma / \sqrt{N}$ is the standard error on the mean, $\sigma_{\bar{x}}$.

Reminder: $V[\bar{x}] = V[X]/N = \sigma^2/N \rightarrow \sigma_{\bar{x}} = \sigma / \sqrt{N}$.

The test statistic z is standard normal, so we use `from scipy.stats import norm` for probabilities.

Example of the Z Test

Is the mean height of DAT students equal to the mean UK height?

True Variance is known to be $\sigma^2 = 50$ cm. The Null hypothesis is $H_0 : \mu_0 = 170$ cm.

- $\alpha = 0.05$ The significance level, chosen before looking at the data.
- $H_1 : \mu \neq 170$ cm The Alternate Hypothesis. ⇨ two-sided test.
- Statistics from data: $N = 57$, $\bar{x} = 167$ cm.

$$z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{N}} = \frac{167 - 170}{\sqrt{50/57}} = -3.203$$

The p-value is `2 * norm.sf(3.203) = 0.00136` ⇨ we reject H_0 , because $p < \alpha$.

Note that `norm.sf(|z|) = 1 - norm.cdf(|z|) = norm.cdf(-1*|z|)`

The (Student's)* T Test

Example usage: test if the sample mean $\bar{x}$ equals a hypothesised mean μ_0 .

Restrictions: Data are approximately $X \sim \text{Norm}(\mu, \sigma)$.

$$T = \frac{\bar{x} - \mu_0}{\sigma_s / \sqrt{N}}$$

The **sample variance** used in the T Test is an unbiased estimator for the true variance: $\sigma_s^2 = \frac{1}{N-1} \sum_i^N (x_i - \bar{x})^2$.

T has its own PDF, so we use `from scipy.stats import t` for probabilities.

***Student** was the pen name of William Gosset. [\[wikipedia\]](#): "Gosset was a friend of both Pearson (χ^2) and Fisher (I_θ), a noteworthy achievement, for each had a massive ego and a loathing for the other."

Example of the T Test

Same question: Is the mean height of DAT students equal to the mean UK height?

As for the Z Test, $H_0 : \mu_0 = 170 \text{ cm}$, $\alpha = 0.05$, $H_1 : \mu \neq 170 \text{ cm}$ ⇨ two-sided test.

■ Statistics from data: $N = 57$, $\bar{x} = 167 \text{ cm}$, $\sigma_s = 10 \text{ cm}$.

$$T = \frac{\bar{x} - \mu_0}{\sigma_s / \sqrt{N}} = \frac{167 - 170}{10 / \sqrt{57}} = -2.265$$

The p-value is $2 * \text{t.sf}(2.265, \text{df}=57-1) = 0.02740$ ⇨ we reject H_0 , as $p < \alpha$.

$\text{df} = 57-1$ is the Number of Degrees of Freedom, $\nu = N - N_p - N_c$.

Note that $\text{t.sf}(|T|, \text{df}) = 1 - \text{t.cdf}(|T|, \text{df}) = \text{t.cdf}(-1 * |T|, \text{df})$

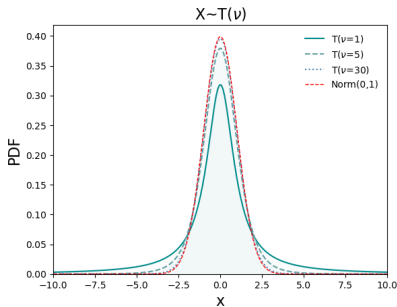
Compare Z and T Tests

Z test found p-value `2*norm.sf(3.203)= 0.00136` using true $\sigma = 7$ cm.

T test found p-value `2*t.sf(2.265, df=57-1)= 0.02740` using sample $\sigma_s = 10$ cm.

Two things are happening here: the σ are different, and the T distribution is not quite normal, it has fatter tails.

Note that using σ_s for the Z Test, the p-value `norm.sf(2.265)=0.0235` is smaller than for the T test.



Two Sample T Test

Example usage: test if two samples have the same mean.

Restrictions: Data are $X \sim \text{Norm}(\mu, \sigma)$. Roughly equal variances ($0.5 < \frac{\sigma_x}{\sigma_y} < 2$).

$$t = \frac{\bar{x} - \bar{y}}{\sigma_p}$$

The **Pooled Variance:** $\sigma_p^2 = \frac{\nu_x \sigma_{sx}^2 + \nu_y \sigma_{sy}^2}{\nu_x + \nu_y}$

Here ν_x is the number of degrees of freedom for sample X ($\nu = N - 1$), and σ_{sx}^2 is the variance of sample X.

Example of the Two Sample T Test

Are the mean heights of DAT students and Mechanics student equal?

The Null hypothesis is H_0 : they are equal. We keep $\alpha = 0.05$.

The Alternate hypothesis is H_1 : DAT are taller ⇨ **one-sided test**.

■ DAT heights: $N = 57$, $\bar{x} = 167$ cm. $\sigma_x = 10$ cm.

■ Mechanics heights: $N = 53$, $\bar{x} = 165$ cm. $\sigma_x = 11$ cm.

$$\sigma_p^2 = \frac{\nu_x \sigma_{sx}^2 + \nu_y \sigma_{sy}^2}{\nu_x + \nu_y} = \frac{(57-1)10^2 + (53-1)11^2}{(57-1) + (53-1)} = 110$$

$$t = \frac{\bar{x} - \bar{y}}{\sigma_p} = (167 - 165) / \sqrt{110} = 0.191$$

The p-value is `t.sf(0.191, df=108) = 0.424` ⇨ we do not reject H_0 , because $p > \alpha$.

We could perhaps have guessed this. T is roughly normal, and for $p < 0.05$ we would need the means to be about 1.6 sigmas away from one another.

The F Test (One-way ANOVA)

Another way to compare groups is via ANalysis Of VAriance. The F test is the ratio:

$$F = \frac{\text{Variance between groups}}{\text{Variance within groups}}$$

- Z is the number of groups j , and N is the number in each group i .

- Mean of group j : $\bar{x}_j = \frac{1}{N} \sum_i^N x_{ij}$.

- Variance of group j : $\sigma_{sj}^2 = \frac{1}{N-1} \sum_i^N (x_{ij} - \bar{x}_j)^2$.

- Mean across all groups: $\bar{x} = \frac{1}{Z} \sum_j^Z \bar{x}_j$.

$$F = \frac{\frac{1}{Z-1} \sum_j^Z N_j (\bar{x}_j - \bar{x})^2}{\frac{1}{Z} \sum_j^Z \sigma_{sj}^2}$$

F has its own PDF, so we use `from scipy.stats import f` for probabilities.

Example of the F Test

Are the mean heights of DAT students and Mechanics student equal?

- DAT heights: $N = 57$, $\bar{x} = 167$ cm. $\sigma_x = 10$ cm.
- Mechanics heights: $N = 53$, $\bar{x} = 165$ cm. $\sigma_x = 11$ cm.
- Variance between: $\frac{1}{Z-1} \sum_j N_j (\bar{x}_j - \bar{x})^2 = \frac{1}{2-1} (57(167 - 166)^2 + 53(165 - 166)^2) = 110$
- Variance within: $\frac{1}{Z} \sum_j \sigma_{sj}^2 = \frac{1}{2} (10^2 + 11^2) = 110.5$

$$F = \frac{110}{110.5} = 0.995$$

Degrees of freedom: Variance between has $Z - 1 = 1$, Variance within has $\sum_j N_j - Z = 108$.

The p-value is `f.sf(0.995, 1, 108) = 0.321` ⇔ we do not reject H_0 , because $p > \alpha$.

Chi Squared and the Standard Normal RV

We have seen that χ^2 is the statistic (the Loss Function) we minimise when using the Least Squares method to estimate parameters and their uncertainties.

$$\chi^2 = \sum_i^N \frac{(y_i - \mu)^2}{\sigma_i^2}$$

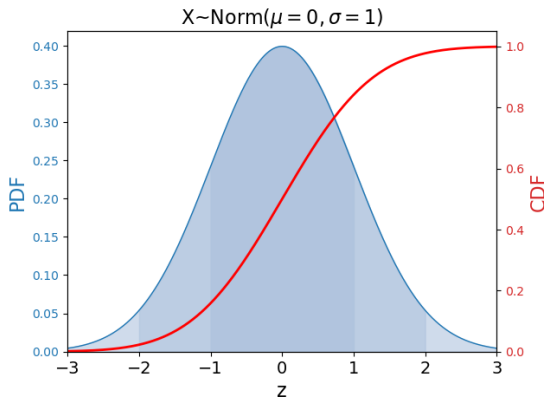
It may have caught your eye that χ^2 is the same form as the Standard Normal RV:

$$Z = \frac{(x - \mu)}{\sigma}$$

For a single measurement (a single **Degree of Freedom**), $\chi^2 = Z^2$.

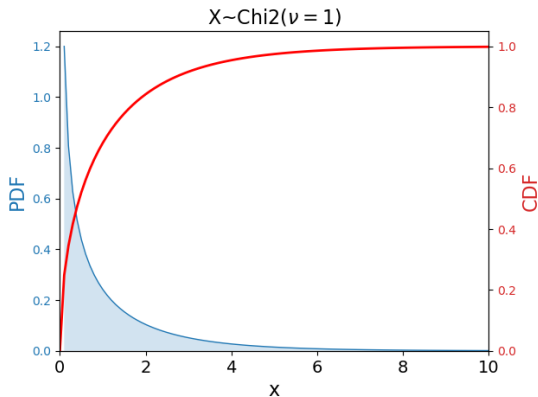
Standard Normal Distribution

The Standard Normal distribution is familiar and welcoming.



Chi Squared Distribution: 1 degree of freedom

The χ^2 distribution looks weird, but makes sense.



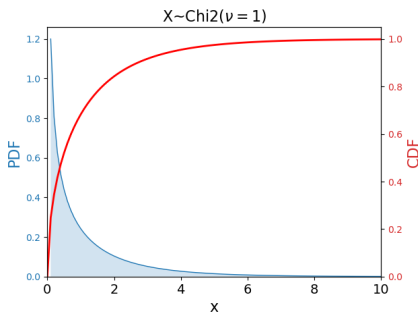
Chi Squared Distribution: 1 degree of freedom

The x-axis is the size of the χ^2 for our single measurement.

$$x = \chi^2(\nu = 1) = z^2 = \frac{(y_i - \mu)^2}{\sigma^2}$$

The **CDF y-axis** is the probability of measuring a residual smaller or equal to that value.

$$P(\chi^2 \leq x) = F_x = \int_0^x f_X dx: \text{ CDF}$$



The tail of the $\text{Chi2}(\nu)$ PDF goes to ∞ (I have truncated).

Chi Squared PDF: a quick look

$$f_X(x; \nu) = \frac{2^{-\nu/2}}{\Gamma(\nu/2)} x^{(\nu/2)-1} e^{-x/2} \quad \text{for } 0 \leq x < \infty$$

Number of Degrees of Freedom: $\nu = N - N_p - N_c$

The **Expected Value** of $X \sim \text{Chi2}(\nu)$ is $E[X] = \nu$, **Variance** is $V[X] = 2\nu$.

The $\Gamma(\nu/2)$ in the denominator is the **Gamma Function**:

$$\Gamma(s) = \int_0^{\infty} x^{s-1} e^{-x} dx$$

The nice properties of this function include $\Gamma(1/2) = \sqrt{\pi}$, $\Gamma(n) = (n-1)!$ where n is a positive integer, and $\Gamma(s+1) = s\Gamma(s)$.

$$\nu = 1 \rightarrow \Gamma(1/2) = \sqrt{\pi}$$

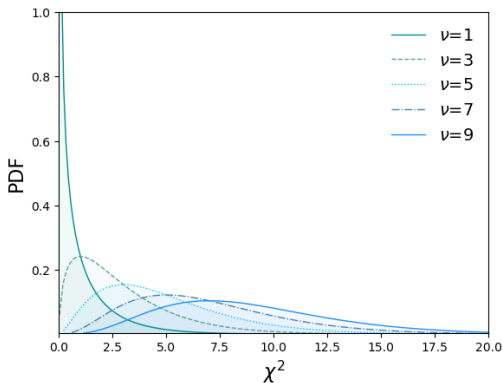
$$\nu = 2 \rightarrow \Gamma(2/2) = 0! = 1$$

$$\nu = 3 \rightarrow \Gamma(3/2) = \Gamma(1/2 + 1) = \frac{1}{2}\sqrt{\pi}$$

and so on.

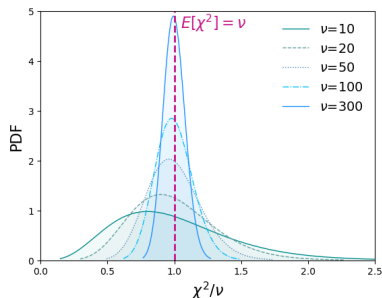
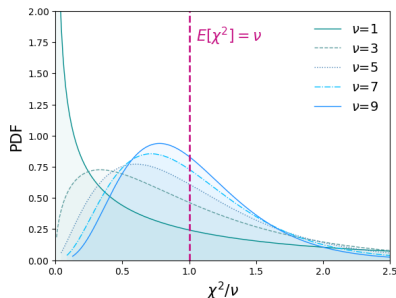
Chi Squared Distribution

When we increase the degrees of freedom, the probability distribution for χ^2 completely changes, because χ^2 is the sum of the squared residuals.



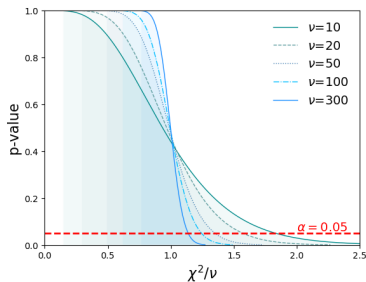
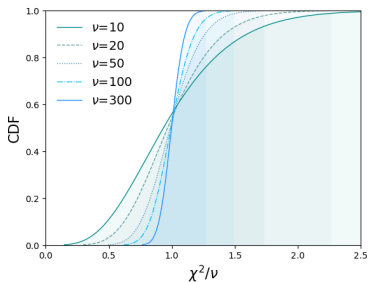
Reduced Chi Squared (χ^2/ν) Distribution

Dividing the χ^2 by its Expected Value, $E[\chi^2] = \nu$, standardises it. As we increase ν , the χ^2/ν distribution approaches $\text{Norm}(\mu = 1, \sigma = \sqrt{2/\nu})$.



Reduced Chi Squared (χ^2/ν) Probability

The reduced χ^2/ν p-value (1-CDF) allow us to view our fit as a hypothesis, and reject it if it falls below some predefined **significance level α** .



Reduced Chi Squared (χ^2/ν) Probabilities

The χ^2/ν should be close to 1. If it is much higher, this indicates a poor fit. If it is much lower, this indicates the uncertainty estimates are too large (overfitting).

You should state your fit results as eg $\chi^2/\nu = 15/10$ and NOT as eg $\chi^2/\nu = 1.5$.

$$P(\chi^2 \geq 15 | \nu = 10) :$$

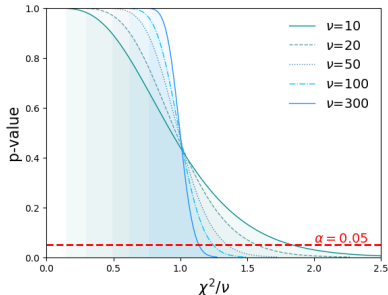
```
chi2.sf(15,10)=0.132
```

okay

$$P(\chi^2 \geq 150 | \nu = 100) :$$

```
chi2.sf(15,10)=0.001
```

excluded!



Likelihood Ratio Test for Goodness of fit

The Log Likelihood Ratio Test is $LLR = -2 \ln \frac{\mathcal{L}_A}{\mathcal{L}_B} = -2(\ln \mathcal{L}_A - \ln \mathcal{L}_B)$.

To use this method, $\mathcal{L}_A$ must be a simplified version of $\mathcal{L}_B$.

Example 1: $\mathcal{L} = \prod_i^N f_{X,i}$, where $f_{X,A} = ax^3$ and $f_{X,B} = ax^3/(1+bx)$. Model A has $N_p = 1$ parameters, Model B has $N_p = 2$. Model A is nested in Model B, because if $b = 0$, they are the same. The difference in ν is $|\nu_A - \nu_B| = 1$.

Example 2: $f_{X,A} = ax^3$, $0.1 < a < 0.2$ and $f_{X,B} = ax^3$, $0 < a < 1$. A has an additional constraint with respect to model B. $|\nu_A - \nu_B| = 1$

We could then use `pvalue=chi2.sf(LLR, df=1)`: if $pvalue < \alpha$, we reject the simpler model, A, in favour of its more complicated friend, B. But, there are other ways of doing this with LLR - more next week.

Comparing Un-nested models: AIC and BIC

If we want to compare un-nested models, we can use the AIC or BIC statistics. They both balance the simplicity of the model (k : number of model parameters) with the quality of fit to data.

The Akaike Information Criterion (AIC):

$$\text{AIC} = 2k - 2 \ln \mathcal{L}_{\hat{\theta}}$$

The Bayesian Information Criterion (BIC):

$$\text{BIC} = k \ln N - 2 \ln \mathcal{L}_{\hat{\theta}}$$

A smaller Information Criterion value is better, eg $\text{AIC}_{\alpha} < \text{AIC}_{\beta}$ identifies the preferred model as α .

Akaike Information Criteria Example

Last week we tossed a coin 50 times and it came up heads 27 times.

The Likelihood under Model α ($p=0.5$):

$$\mathcal{L}_\alpha = \frac{50!}{27!23!} 0.5^{27} 0.5^{23} = 0.09596$$

The Likelihood under Model β ($p=27/50$):

$$\mathcal{L}_\beta = \frac{50!}{27!23!} (27/50)^{27} (23/50)^{23} = 0.11263$$

$$\text{AIC}_\alpha = 2k - 2 \ln \mathcal{L}_\alpha = 2 - 2 \ln(0.09596) = 6.6876$$

$$\text{AIC}_\beta = 2k - 2 \ln \mathcal{L}_\beta = 2 - 2 \ln(0.11263) = 6.3673 \quad \leftarrow \text{smaller preferred}$$

AIC probabilities

We identify the minimum: $m = \min(\text{AIC}_\alpha, \text{AIC}_\beta) = \min(6.6876, 6.3673) = 6.3673$

The **Relative Likelihoods** for AIC are:

$$r_\alpha = \exp\left[\frac{1}{2}(m - \text{AIC}_\alpha)\right] = \exp\left[\frac{1}{2}(6.3673 - 6.6876)\right] = 0.852$$

$$r_\beta = \exp\left[\frac{1}{2}(m - \text{AIC}_\beta)\right] = \exp\left[\frac{1}{2}(6.3673 - 6.3673)\right] = 1$$

The **Akaike Weights** are then:

$$p_\alpha = r_\alpha / (r_\alpha + r_\beta) = 0.852 / 1.852 = 0.460$$

$$p_\beta = r_\beta / (r_\alpha + r_\beta) = 1 / 1.852 = 0.540$$

The weight p_β is the probability that $p = 27/50$ is the best of the two options, given the data. Generally, if the difference $|\text{AIC}_{\text{ff}} - \text{AIC}_{\text{fi}}| < 2$ it is classified as **not worth more than a bare mention** - see [Kass 1995].

Thanks For Your Time

Fin.